

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 4

Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2. Learning Outcomes

After completing this experiment students will be able to

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3. Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

4. Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5. LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.

Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6. AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization and GPU training.

Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7. VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.

Advantages

- Excellent feature extraction
- Simple architecture
- High classification accuracy

8. Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogleNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9. GoogleNet (Inception Network)

GoogleNet, proposed by Szegedy *et al.* in 2014, introduced the **Inception Module**, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogleNet simultaneously applies 1×1 , 3×3 , 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

10. ResNet

Increasing CNN depth often causes the **vanishing gradient problem**, making optimization difficult. ResNet solves this issue using **skip (residual) connections**, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

$$F(x) = H(x) - x$$

thus,

$$Output = F(x) + x$$

where

- $F(x)$ = Residual Mapping
- x = Identity Mapping

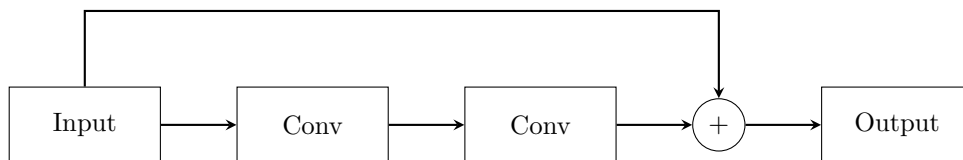


Figure 1: Residual Block in ResNet

Advantages

- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

11. Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters.

The dilation rate determines the spacing between kernel elements.

For a standard convolution,

$$D = 1$$

For dilated convolution,

$$D > 1$$

Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

where zeros indicate inserted gaps.

Applications

- Semantic segmentation
- Medical image analysis
- Satellite imagery
- Object localization

12. Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.

Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13. Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.



Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14. Dataset

Dataset: CIFAR-10

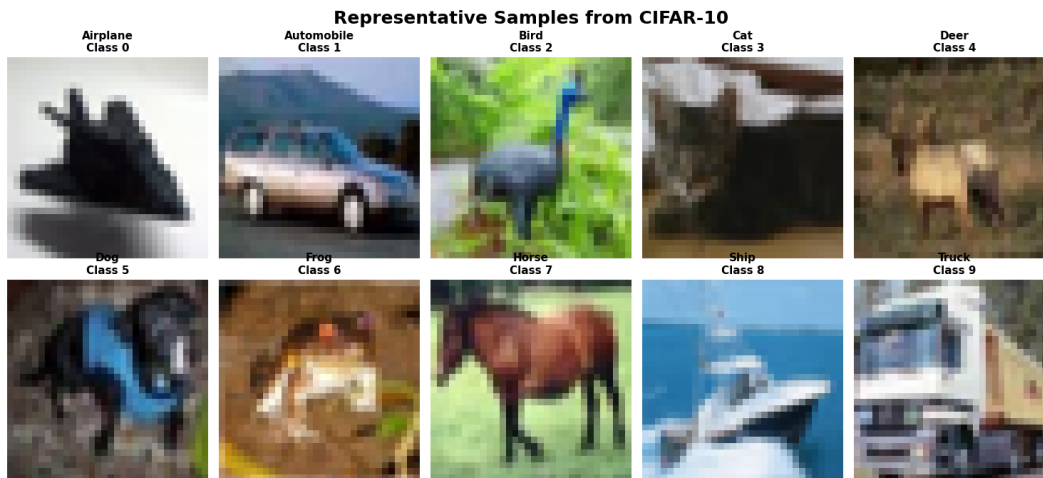
- Training Images : 50,000
- Testing Images : 10,000
- Number of Classes : 10
- Image Size : $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

15. Source Code

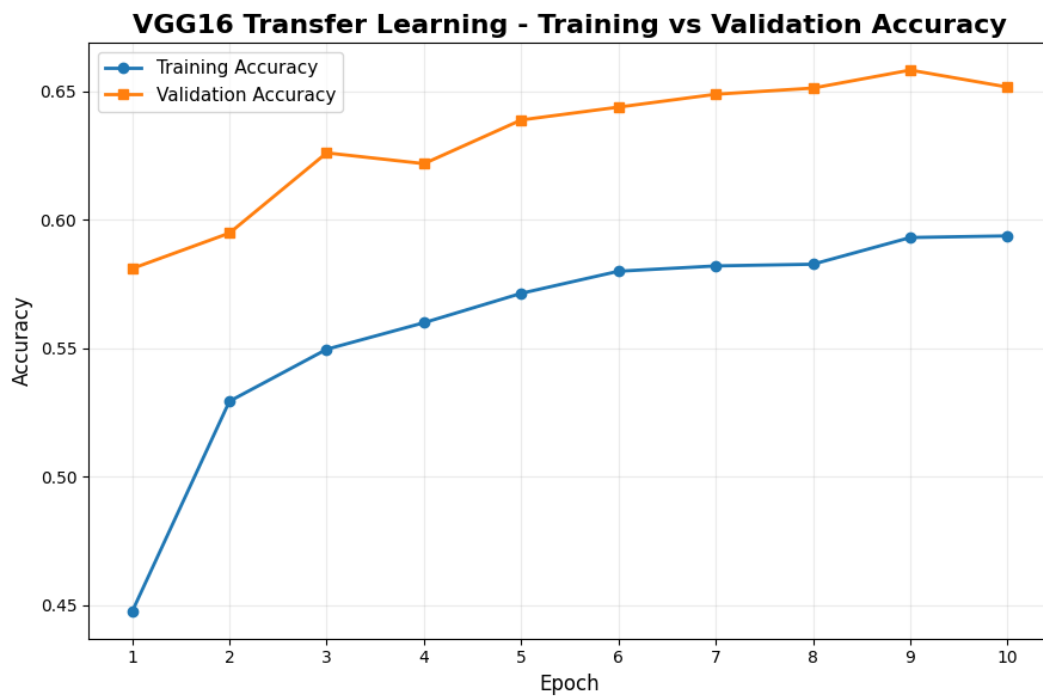
<https://github.com/PranavVarshin/DeepLearning/tree/main/Lab-4>

16.Mandatory Plots

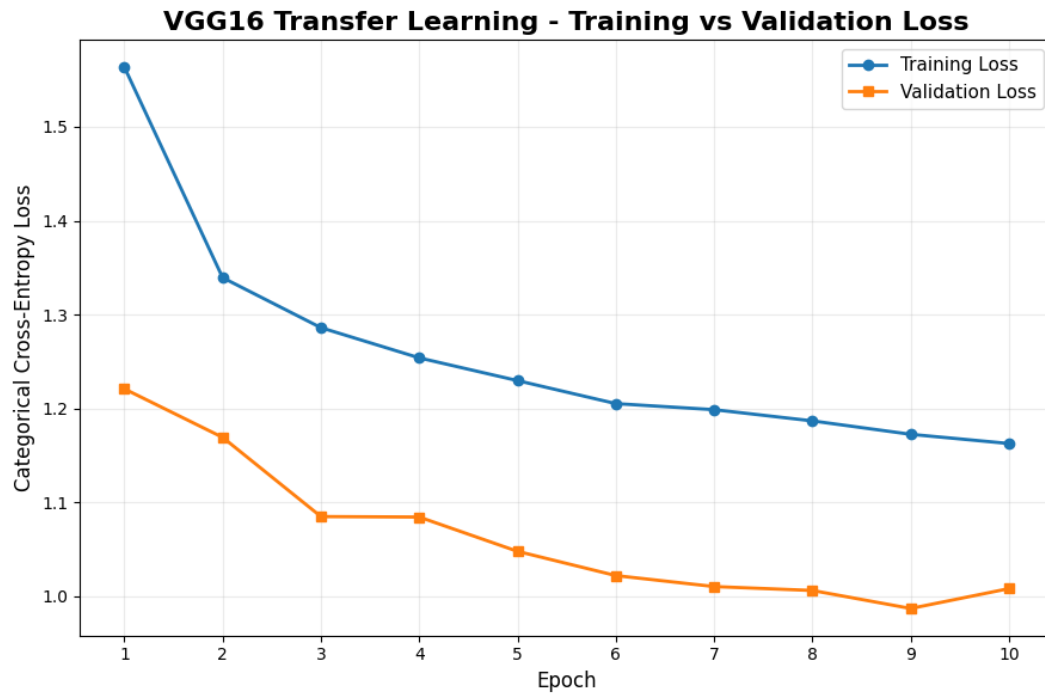
Sample CIFAR-10 images



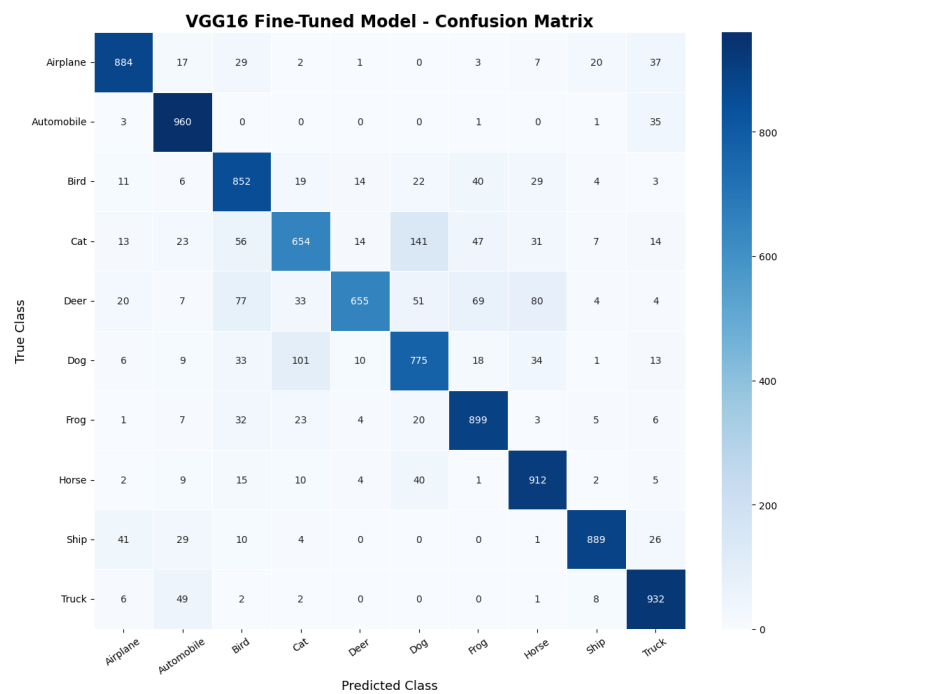
Training Accuracy And Validation Accuracy



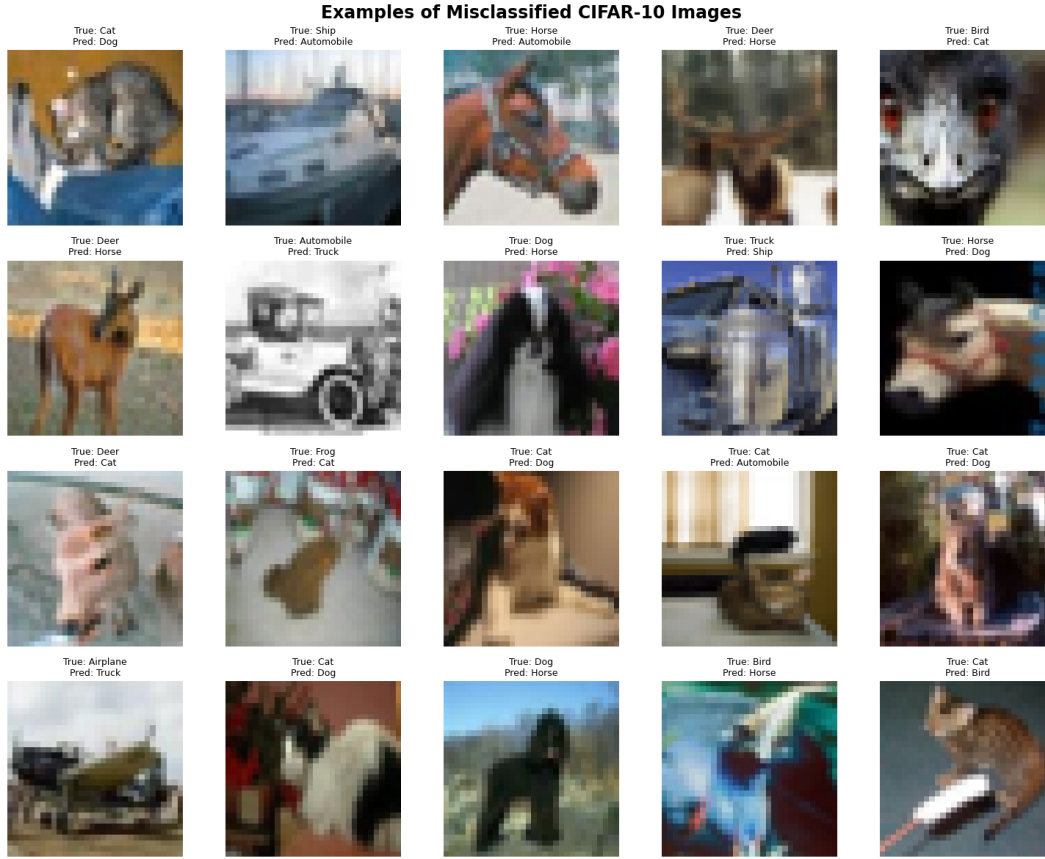
Training Loss and Validation Loss



Confusion Matrix



Missclassified Images



17.Results

Performance Metrics

The performance of the VGG16 transfer learning model after training and fine-tuning was evaluated using accuracy, precision, recall, and F1-score. The total training time includes both the initial training phase and the fine-tuning phase.

Table 1: Performance Metrics of the Proposed Model

Metric	Value
Training Accuracy	65.98%
Testing Accuracy	84.12%
Precision	84.46%
Recall	84.12%
F1-score	83.89%
Training Time	14.67 minutes
Total Parameters	14,848,586

Comparison of CNN Architectures

Different CNN architectures were compared based on the number of parameters, testing accuracy, and training time. LeNet-5, AlexNet, and GoogleNet were trained for three epochs for comparison, while VGG16 and ResNet50 were implemented using transfer learning.

Table 2: Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time
LeNet-5	83,126	47.86	0.36 min
AlexNet	4,483,146	63.18	1.02 min
VGG16	14,848,586	84.12	14.67 min
GoogleNet	27,746	49.30	0.75 min
ResNet50	24,114,826	32.46	3.14 min

The VGG16 transfer learning model achieved the highest testing accuracy of **84.12%**. Fine-tuning improved the testing accuracy from **64.56%** to **84.12%**, resulting in an improvement of **19.56 percentage points**. The results demonstrate that transfer learning and fine-tuning significantly improved the classification performance on the CIFAR-10 dataset.

Among the CNN architectures trained for comparison, AlexNet and the custom ResNet architecture achieved testing accuracies of approximately **63.18%** and **63.45%**, respectively. However, the fine-tuned VGG16 model achieved substantially higher accuracy due to the use of pretrained ImageNet features and fine-tuning of the final convolutional block.

18. Conclusion

In this experiment, different CNN architectures were implemented and compared for CIFAR-10 image classification. Among the evaluated models, the fine-tuned VGG16 model achieved the best testing accuracy of **84.12%**. The experiment demonstrated the importance of CNN architecture design, hyperparameter tuning, transfer learning, and fine-tuning in improving image classification performance.

19. References

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3. Karen Simonyan and Andrew Zisserman, *Very Deep Convolutional Networks for Large-Scale Image Recognition*, ICLR, 2015.
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6. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
7. TensorFlow Documentation: <https://www.tensorflow.org>
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